interdomain routing

so far: routing within one organization

lots of trust/sharing:

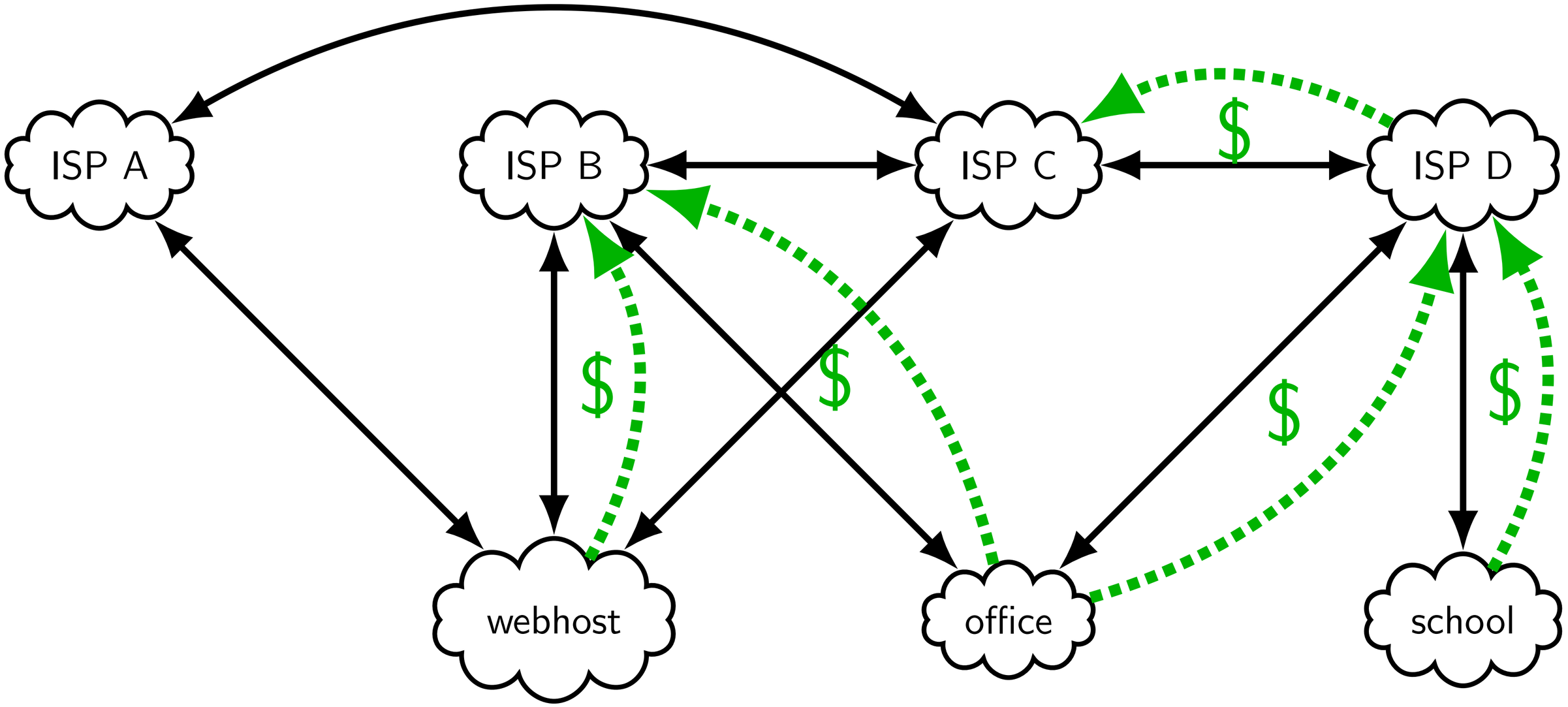
okay to send packets through (essentially) every router

okay for any router to ‘announce’ any address

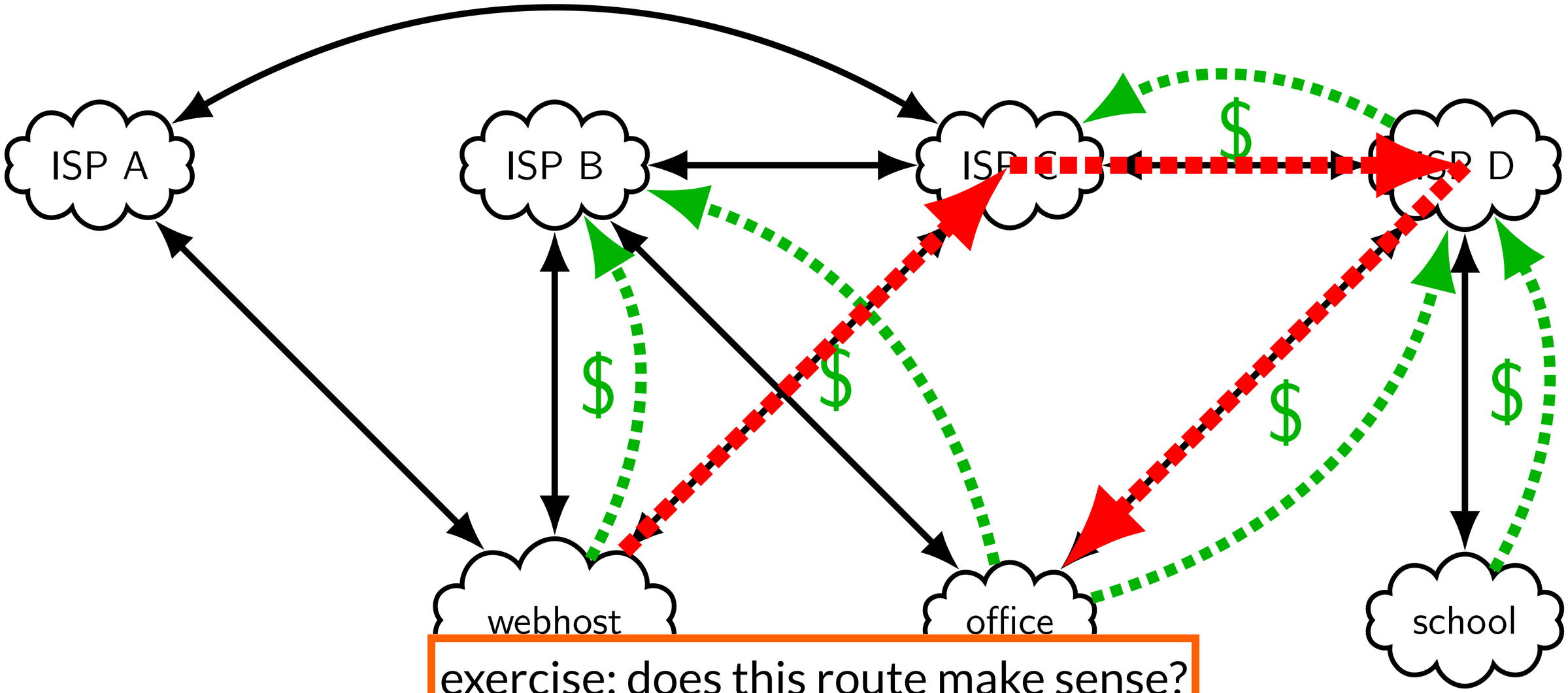
okay to share (almost) full map of network

not what we want for interdomain routing

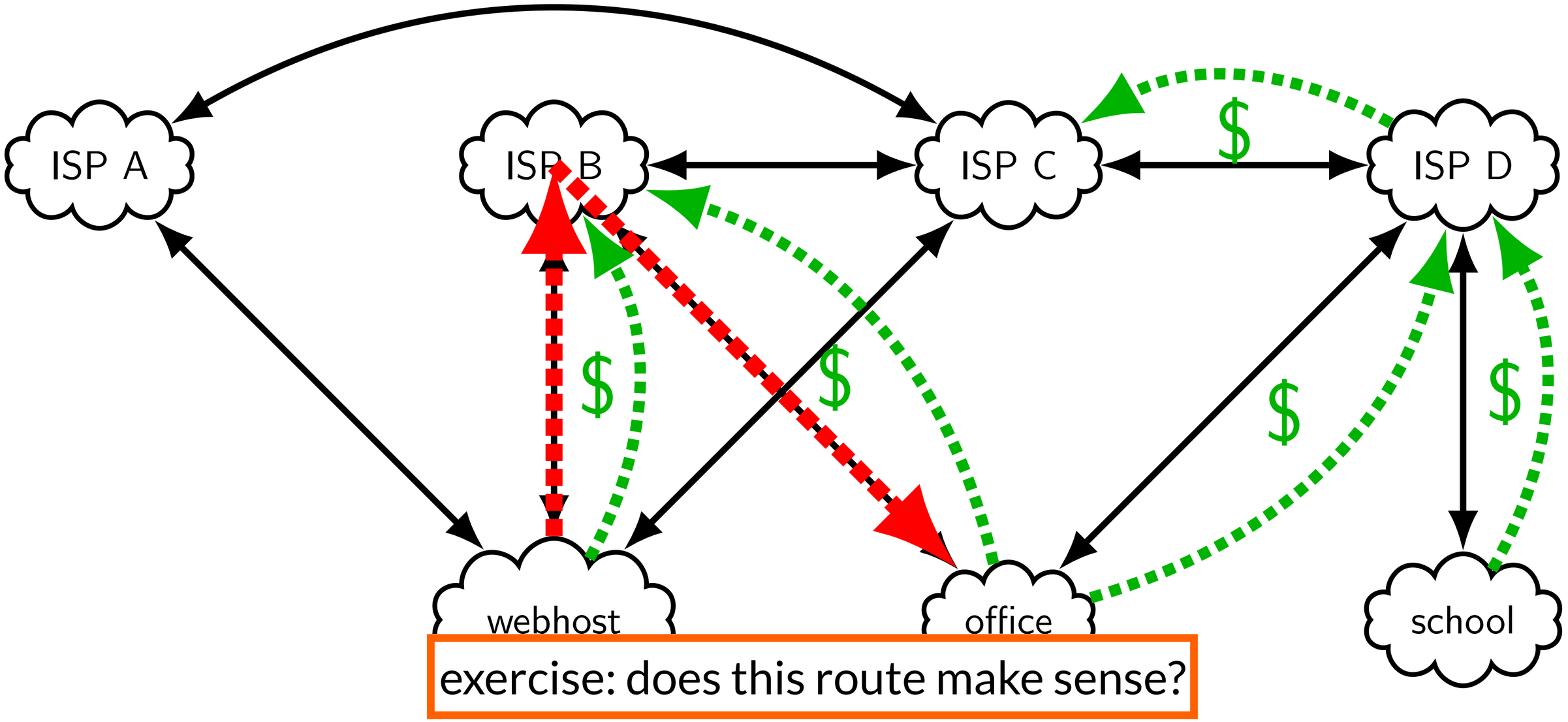
some business considerations



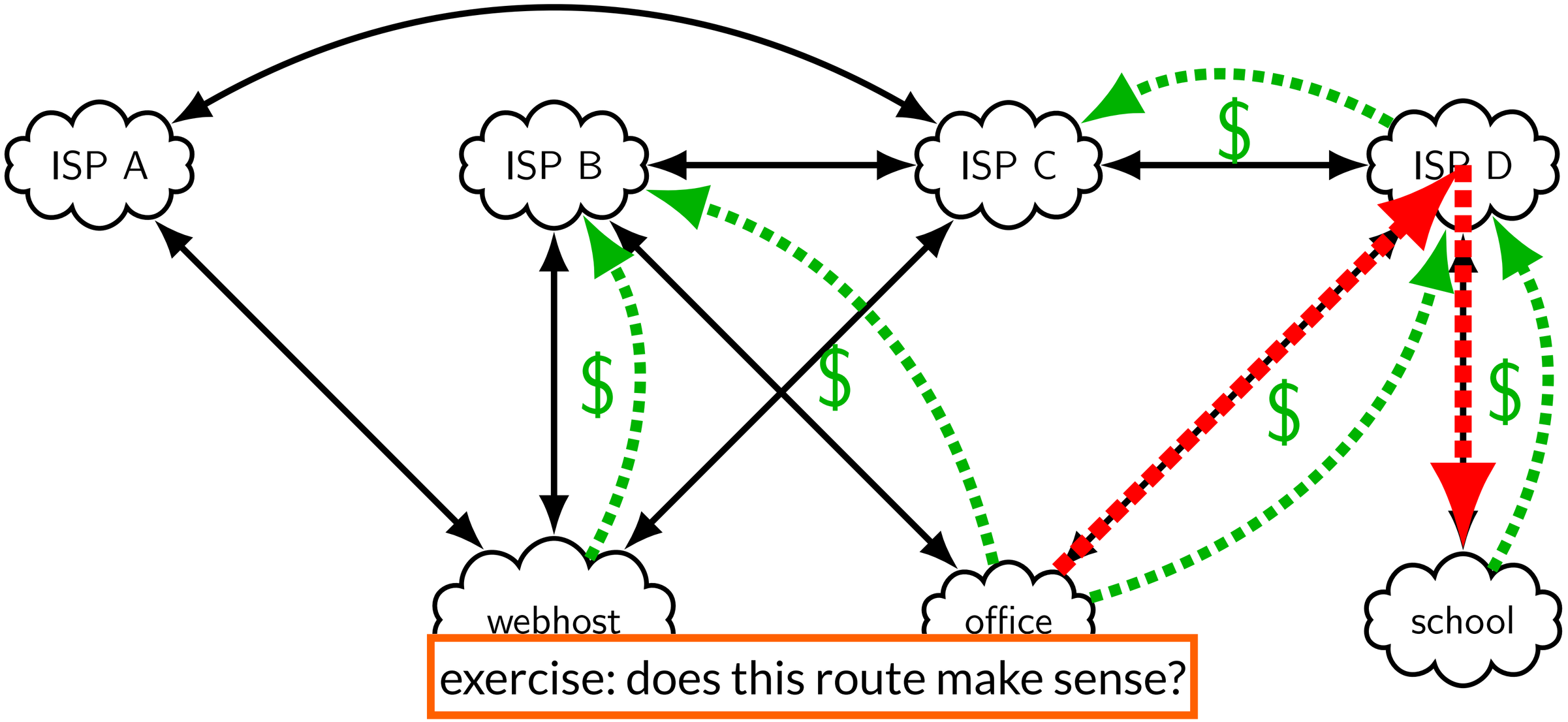
some business considerations (exercise 1)



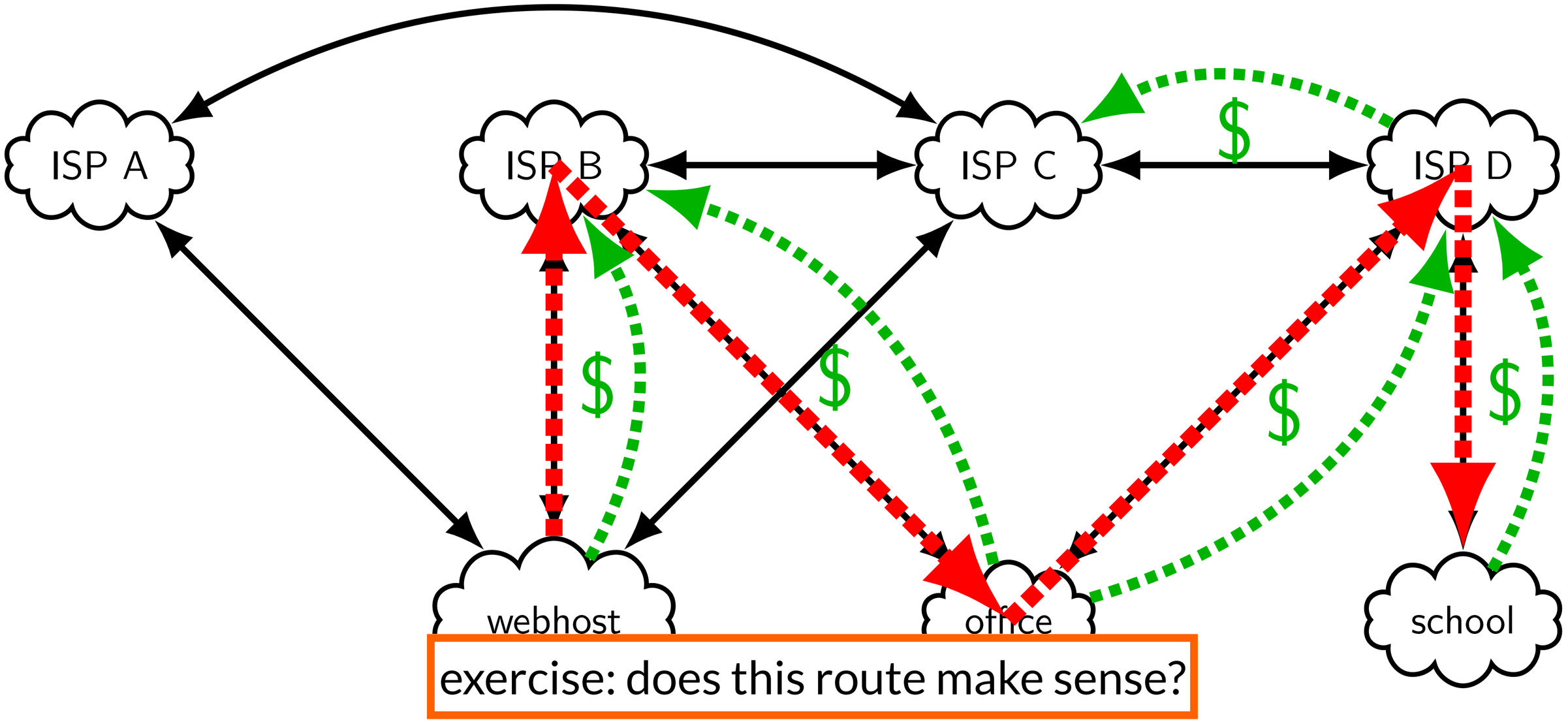
some business considerations (exercise 2)



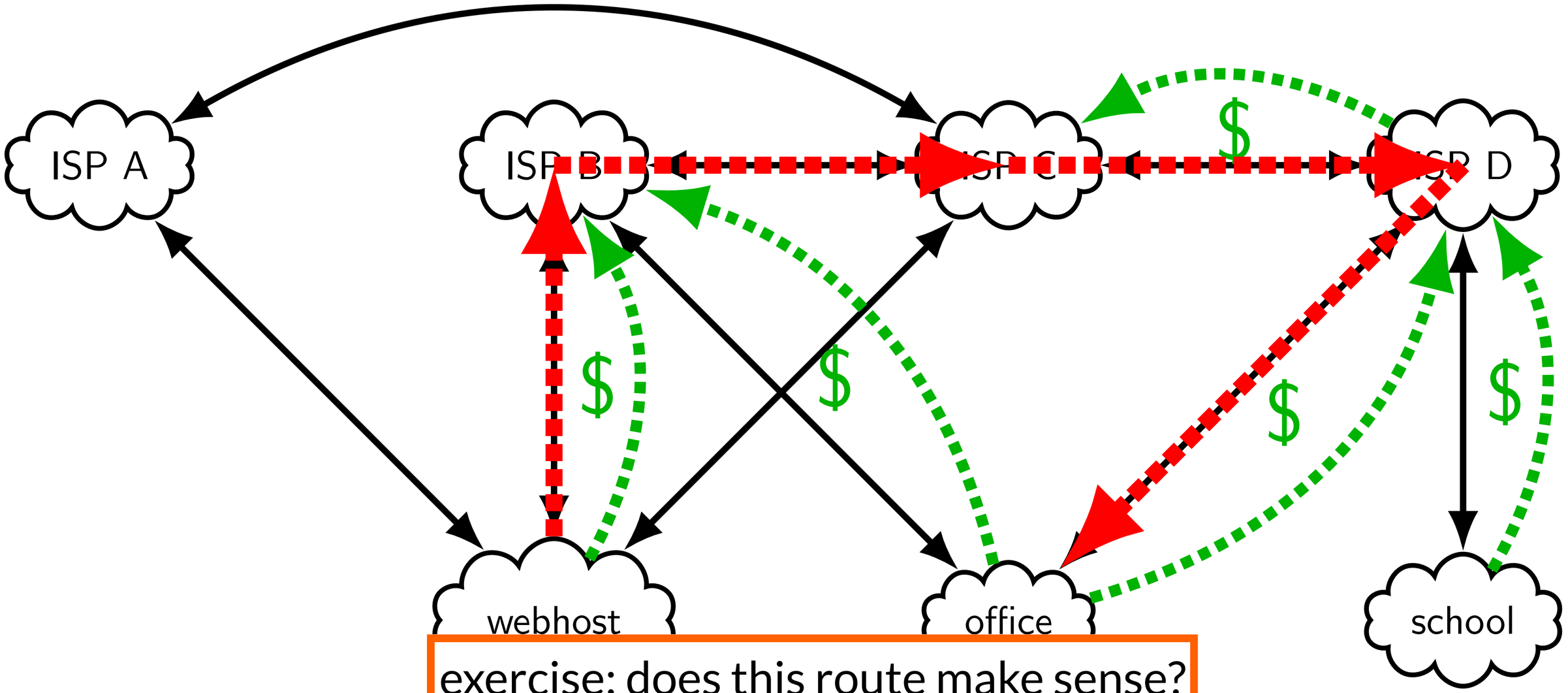
some business considerations (exercise 3)



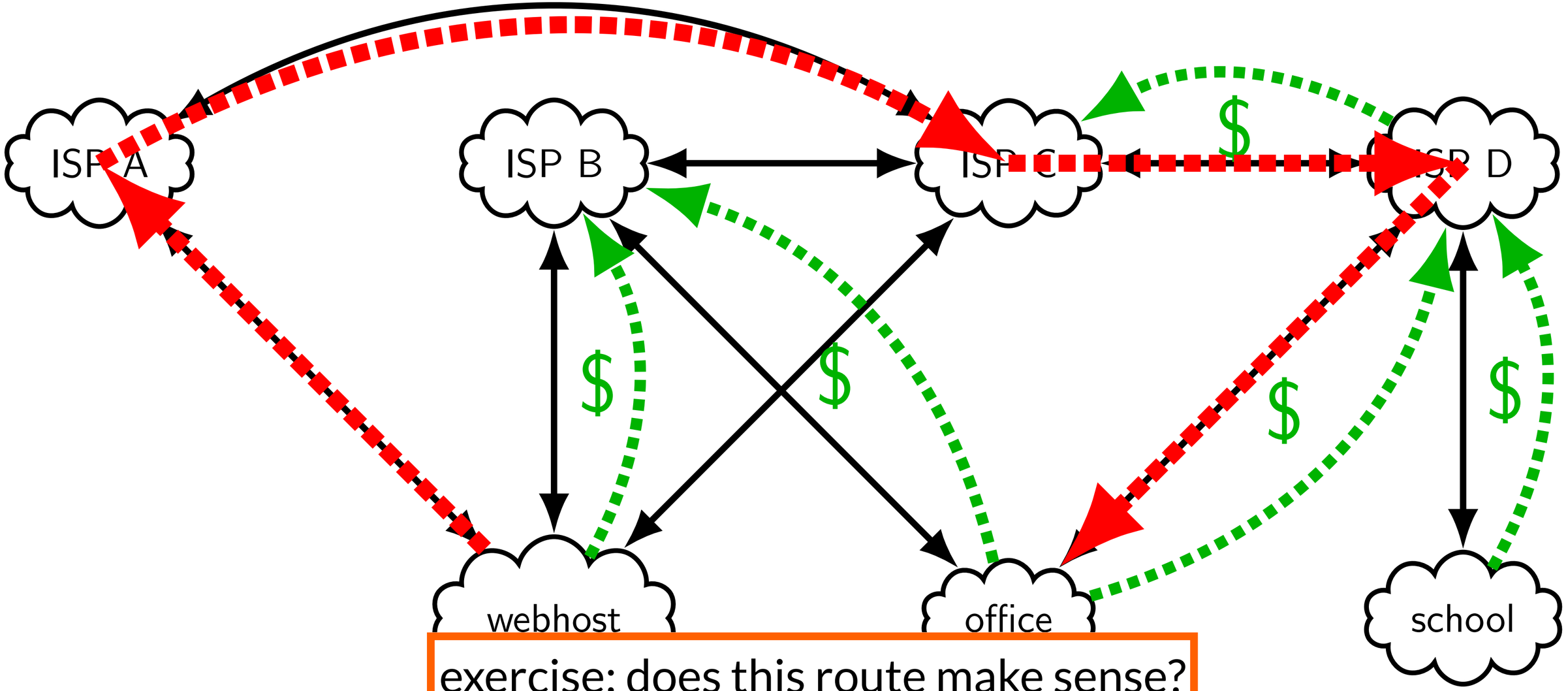
some business considerations (exercise 4)



some business considerations (exercise 5)



some business considerations (exercise 6)



autonomous system

autonomous system (AS) — one “routing domain”

- typically = set of networks administrated by one organization
- decides what routing to use internally
- should be fully connected internally

scope of OSPF instance = one AS

each AS can connect to other ASes

- well-defined protocol for sending routes to other ASes

AS numbers

for Internet routing, ASes are assigned numbers

assigned by IANA and RIRs (similar to IP addresses)

originally 16-bit, now extended to 32-bit

some private use / special AS numbers

relationship types

provider/customer

- typically: customer pays provider
- provider connects customers everywhere it can (customer paid for it)
- customer does *not* provide paths through its network

peer/peer

- often: no payment (‘settlement-free’)
- if A peers with B…
- A gets connected to B’s customers (customers paid B for this)
- A does not get connected to B’s other peers (no one paid B for this)
- A does not get connected to B’s providers (no one paid B for this)

connecting big networks?

some options:

(which are basically the same as connecting parts of big network)

run a fiber between two buildings

permitting and construction needed

pay for direct access to fiber someone else ran (“dark fiber”)

burying one fiber costs similar to burying bundle, so spares

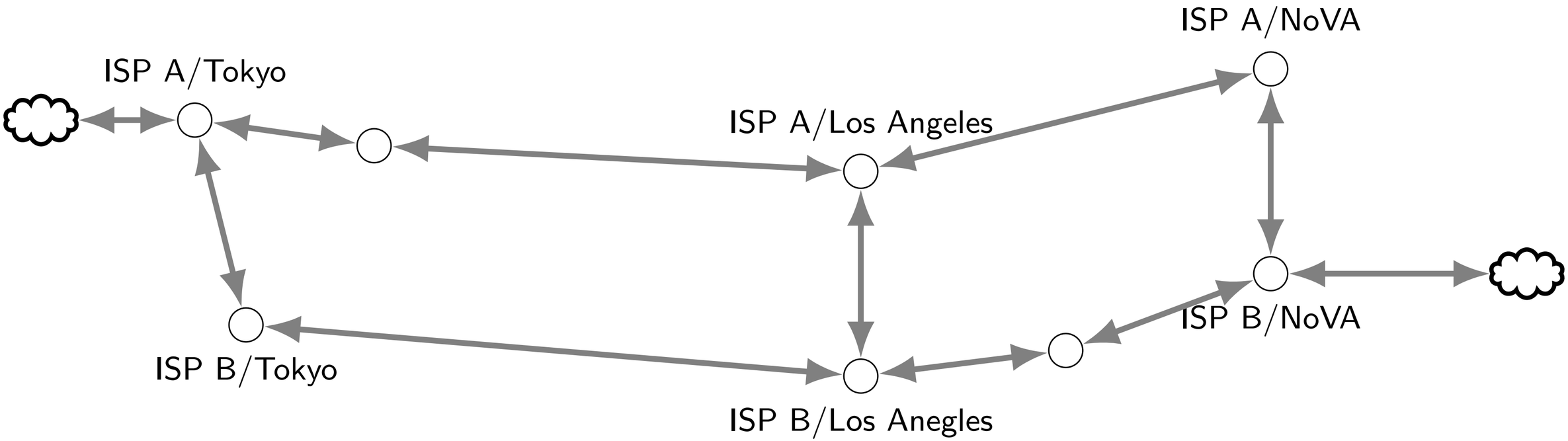
pay a telecom for a site-to-site connection

“gaurenteed” bandwidth+latency between two sites

may or may not use series of dedicated fibers

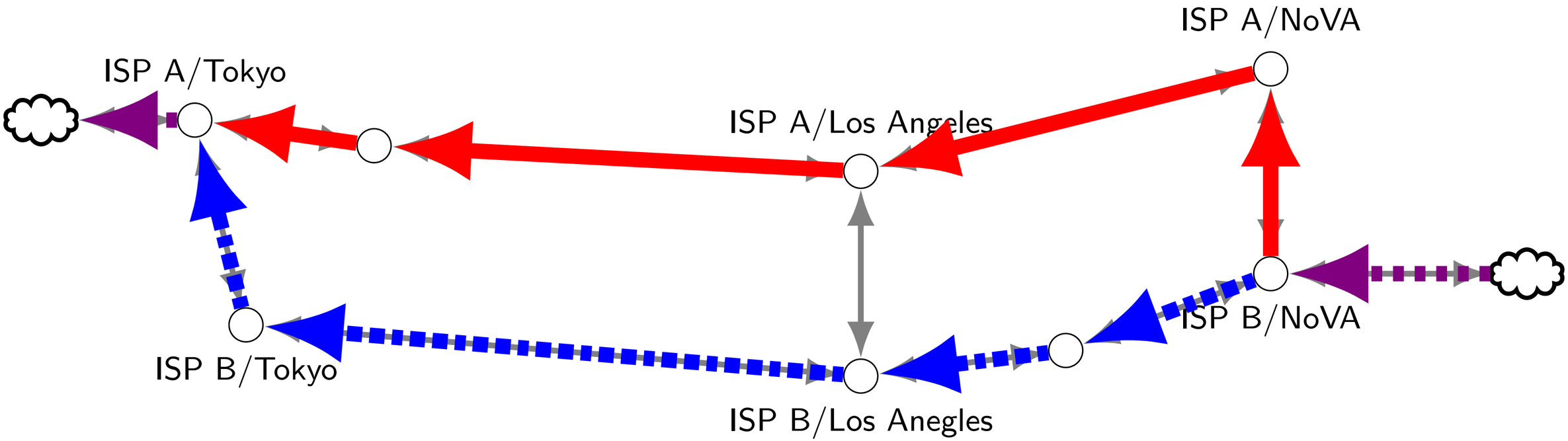
get space in common datacenter, pay datacenter operator for connection

going the distance



does ISP A or ISP B help packets cross the Pacific?

going the distance



does ISP A or ISP B help packets cross the Pacific?

distance preferences

ISP B$\rightarrow$ISP A across the Pacific:

for ISP B:

- cheaper to hand-off packet to ISP A as soon as possible
- more control over performance if handing off as late as possible

for ISP A:

- cheaper to require ISP B to hand-off packet as late as possible
- more control over performance if B sends as soon as possible

maybe part of ISP A and ISP B peering agreement

Border Gateway Protocol

protocol for sending routes between networks

used whereever routers from different ASes connect

“EBGP”

used within AS to share routes out of AS internally

“IBGP”

each router constructs list of routes to offer

each router receives list of routes, exports to OSPF/etc.

BGP connections

BGP (TCP) connections made between routers

each router keeps track of set routes advertised by other

command sent to add or withdraw specific routes

not like distance vector where we kept resending everything

BGP prefixes

routes sent via BGP called ‘prefixes’

said to be “announced” from one router to another

because the network (e.g. 10.0.1.0/24) is the important part

(and the next hop is implied by where prefix comes from)

BGP route

adjacent routers share list of *routes* with:

IP prefix (CIDR-style, basically)

AS path — list of autonomous system the route goes through

next hop router (IP address)

multi-exit discriminator

low value = this entrance to AS is better than others for these IPs

local preference (internal-only)

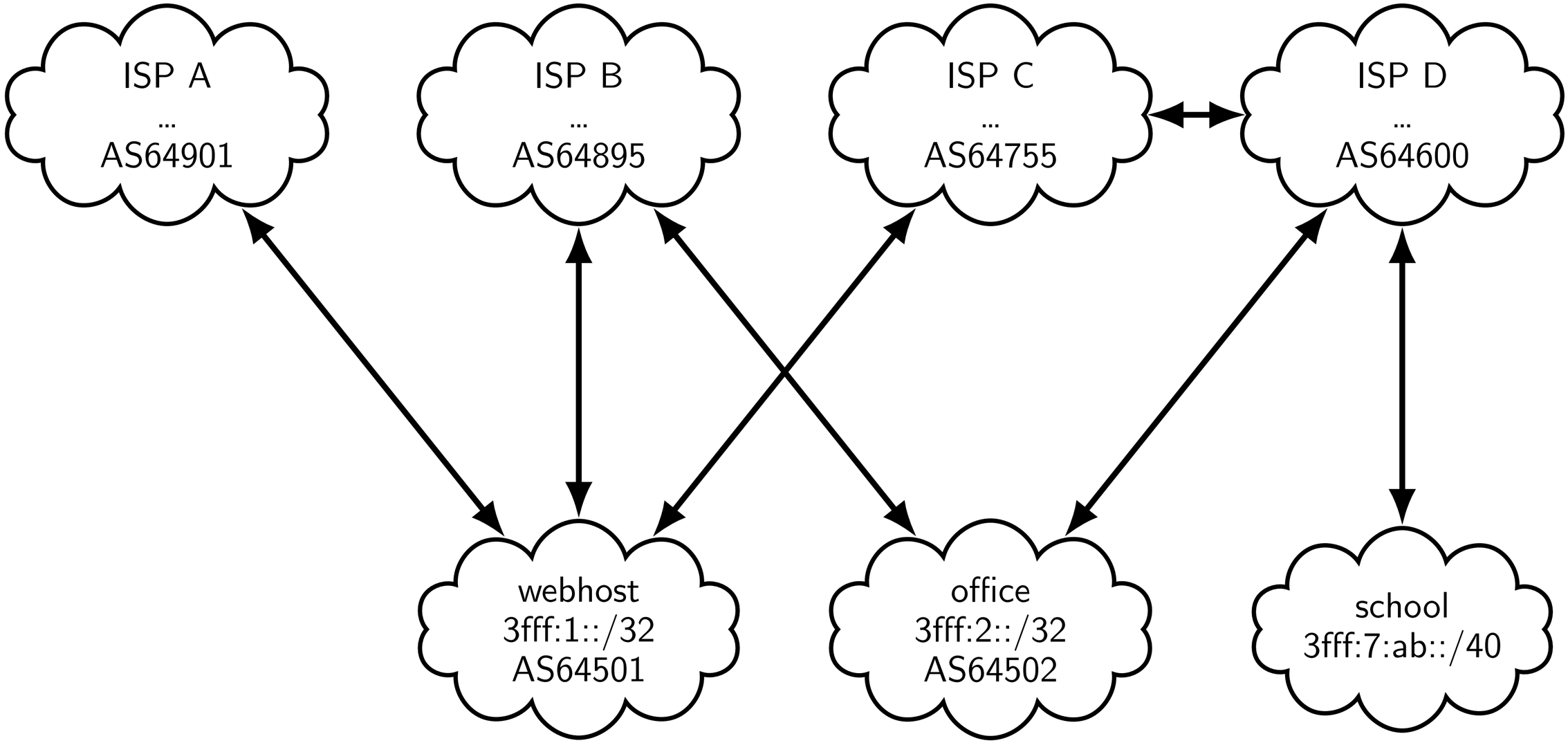
AS path

used to detect routing loops

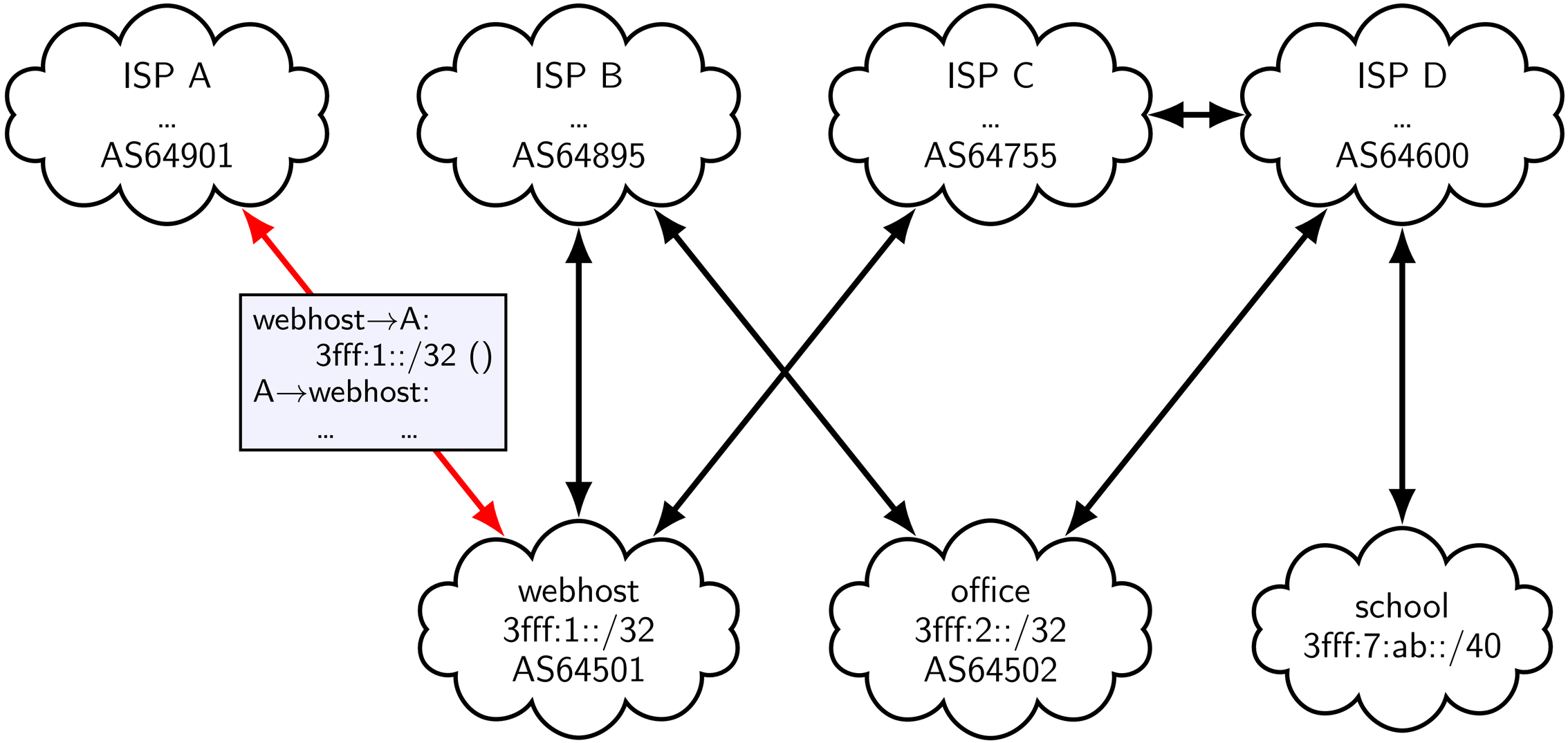
append your AS when sending route externally

always ignore external routes with your AS in their AS path already

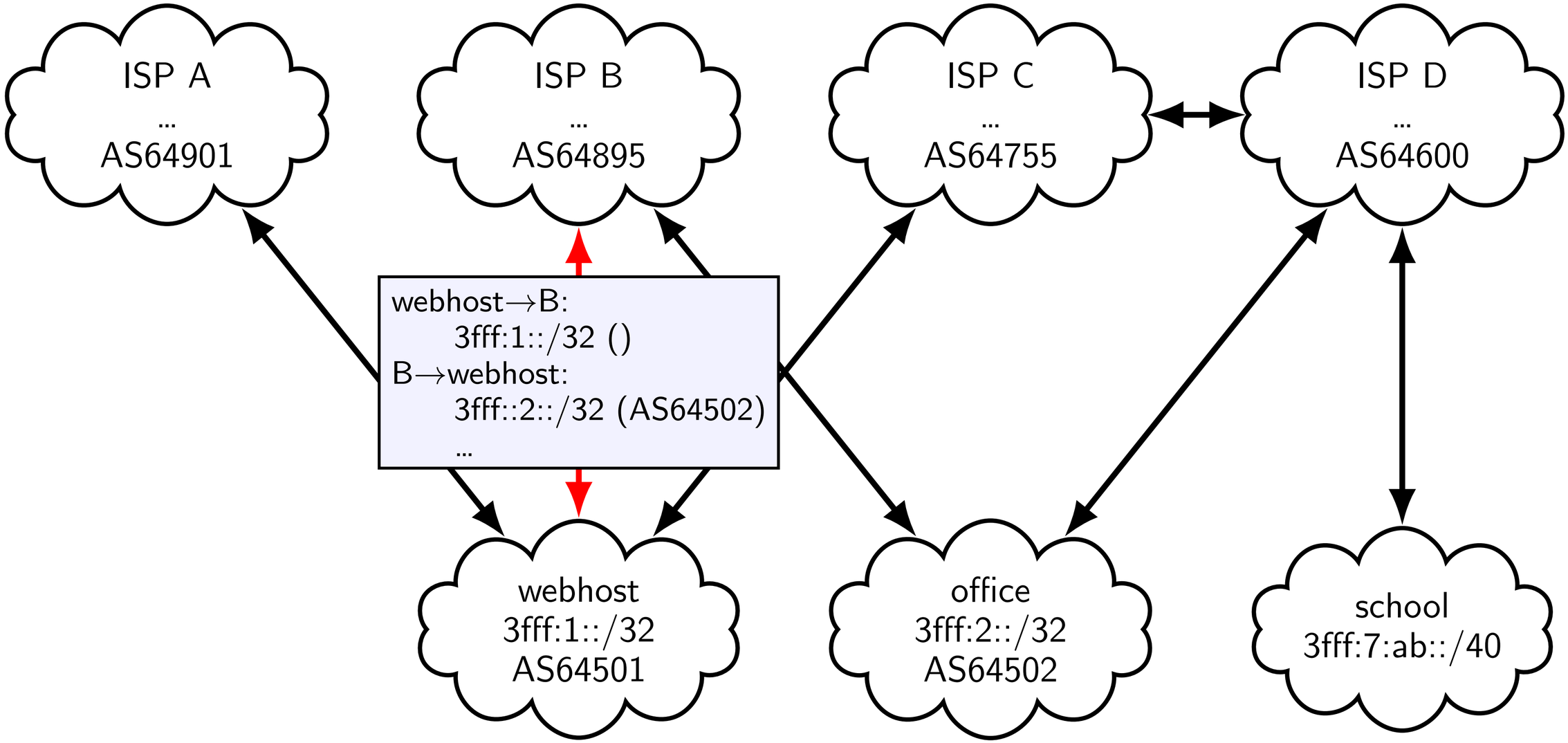
external BGP



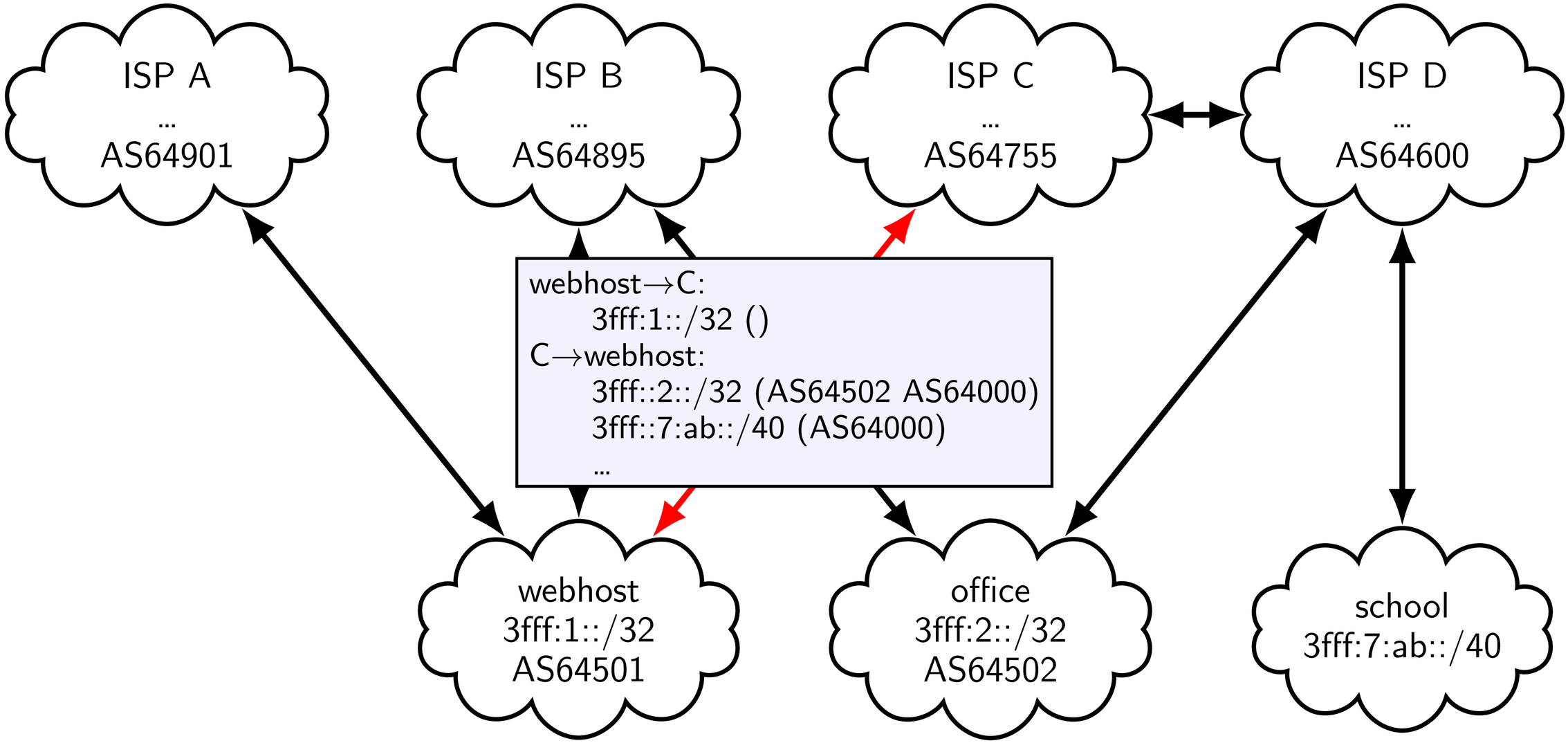
external BGP



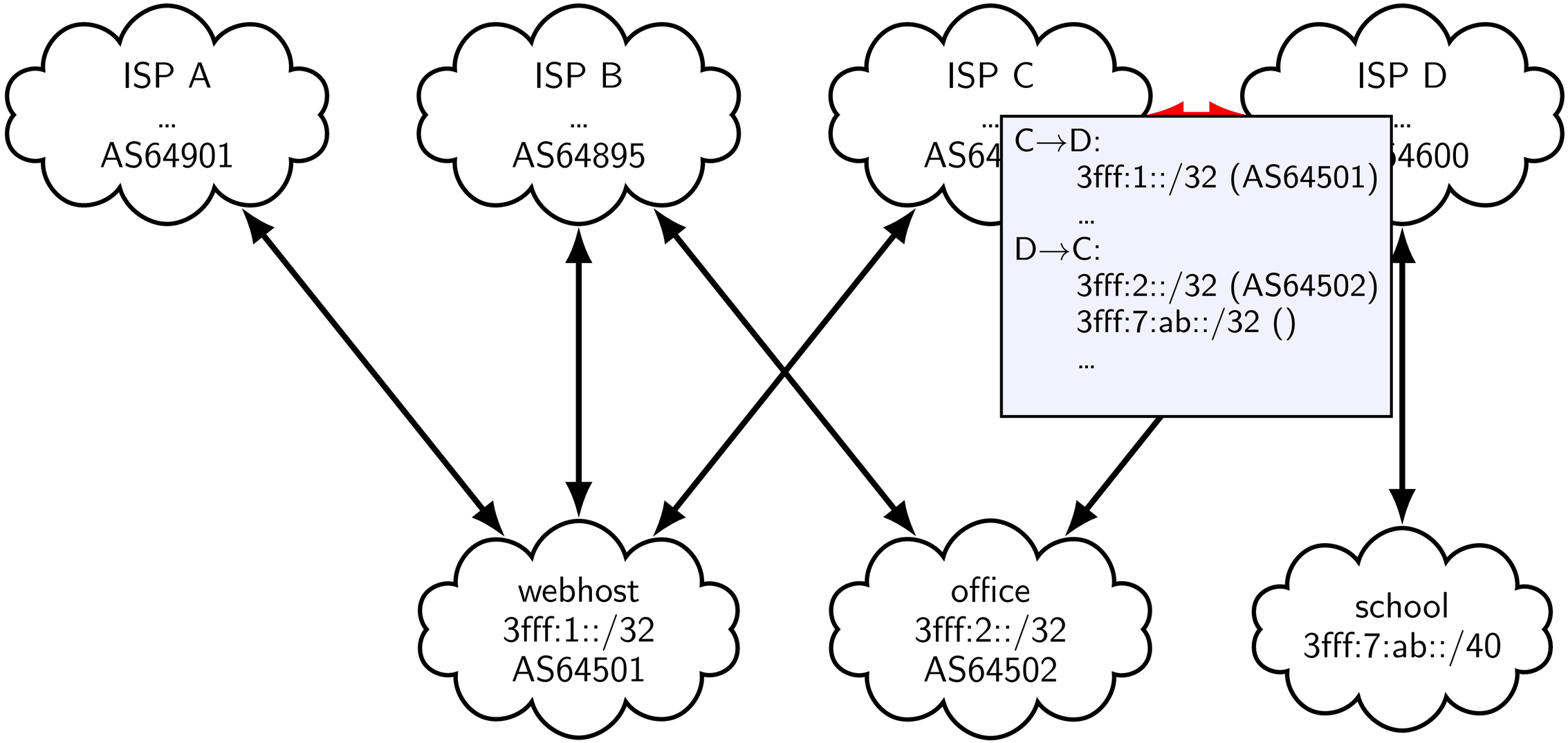
external BGP



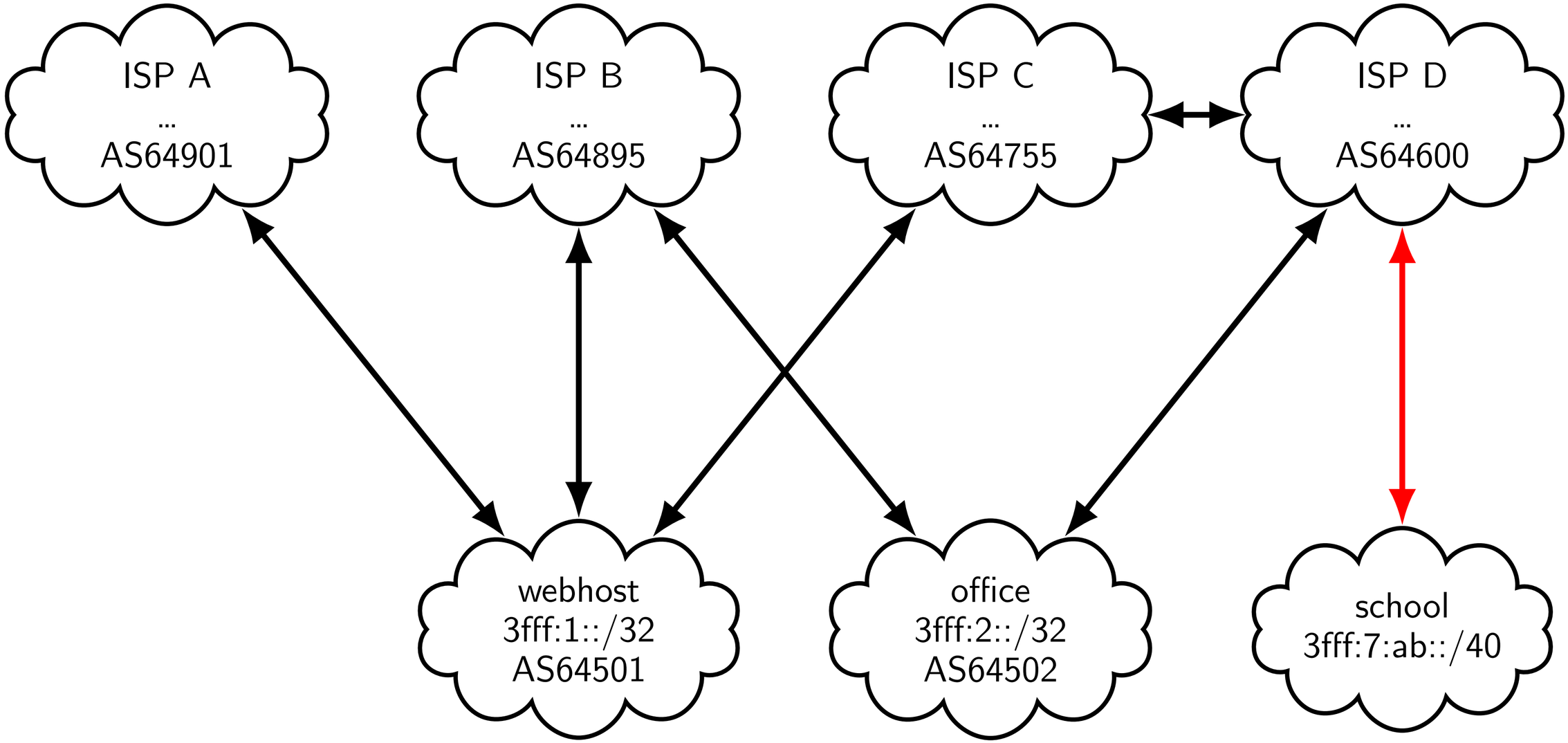
external BGP



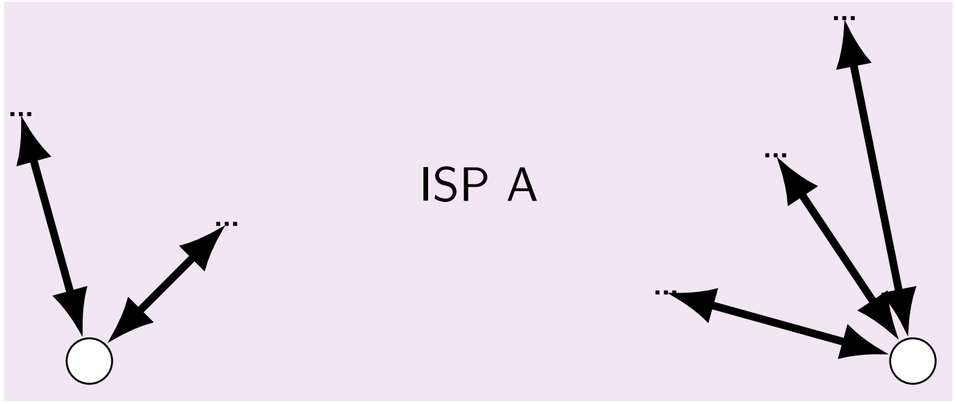
external BGP

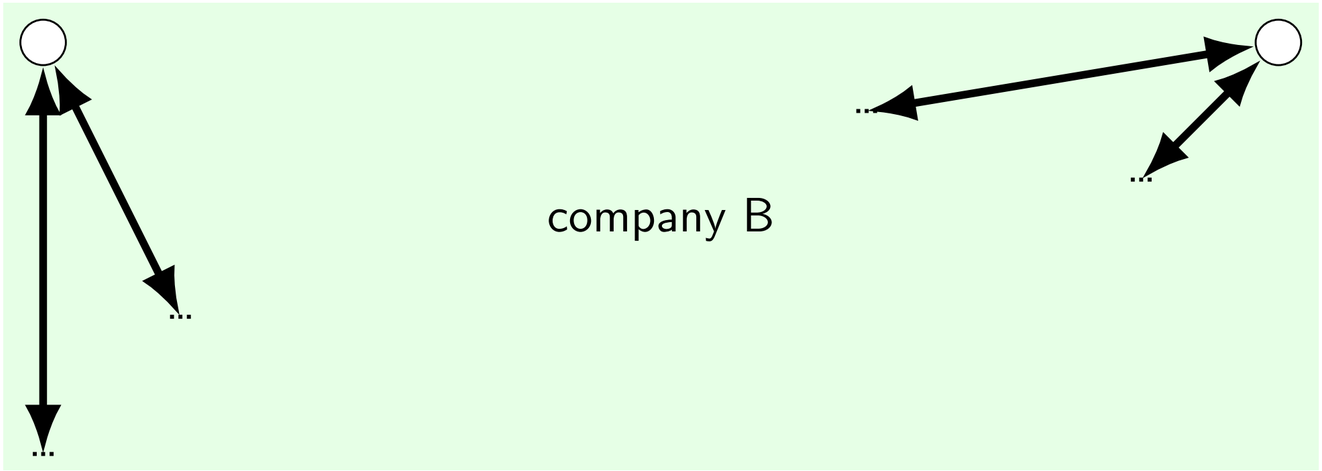


external BGP

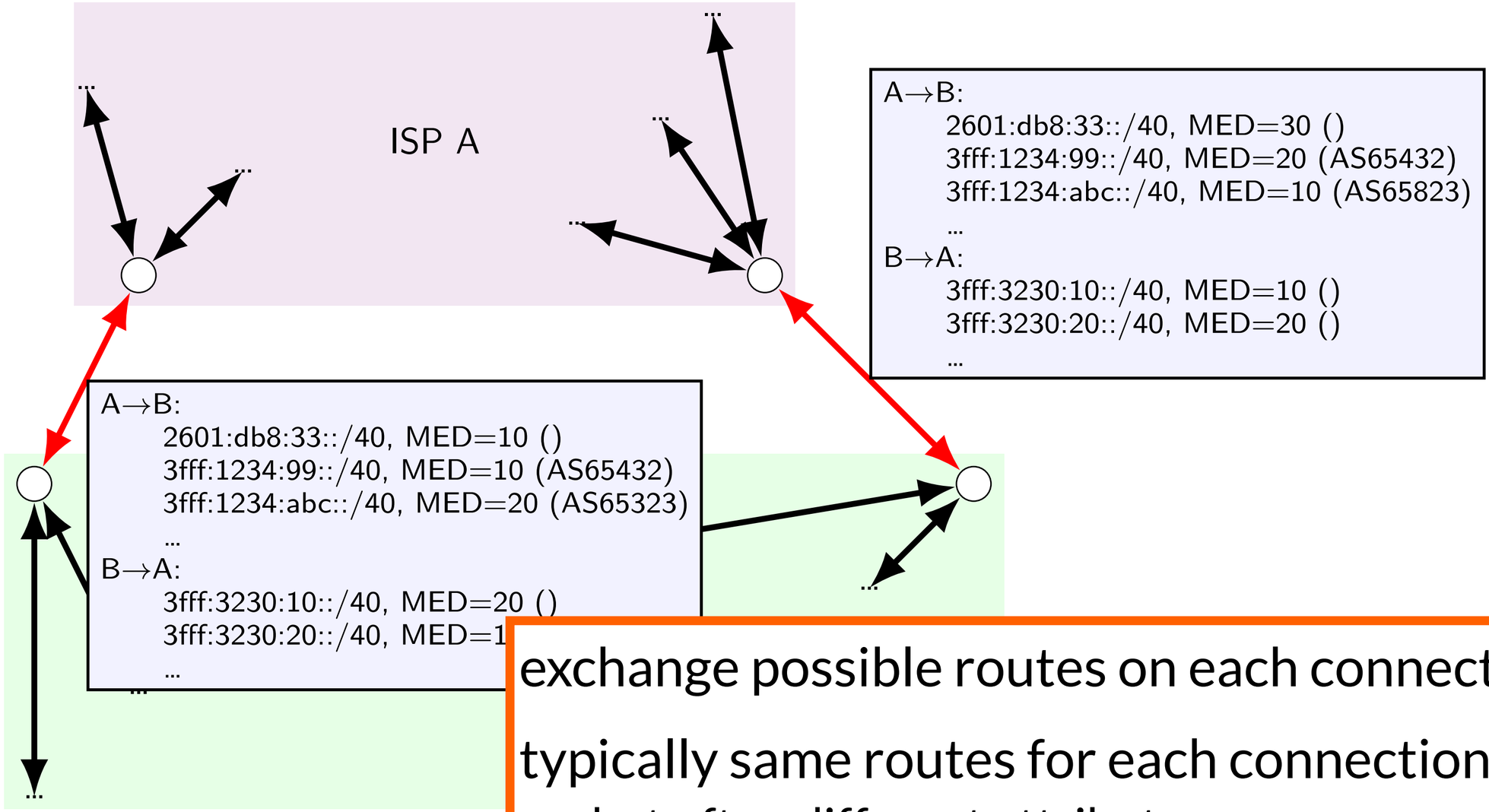


multiple BGP sessions





multiple BGP sessions



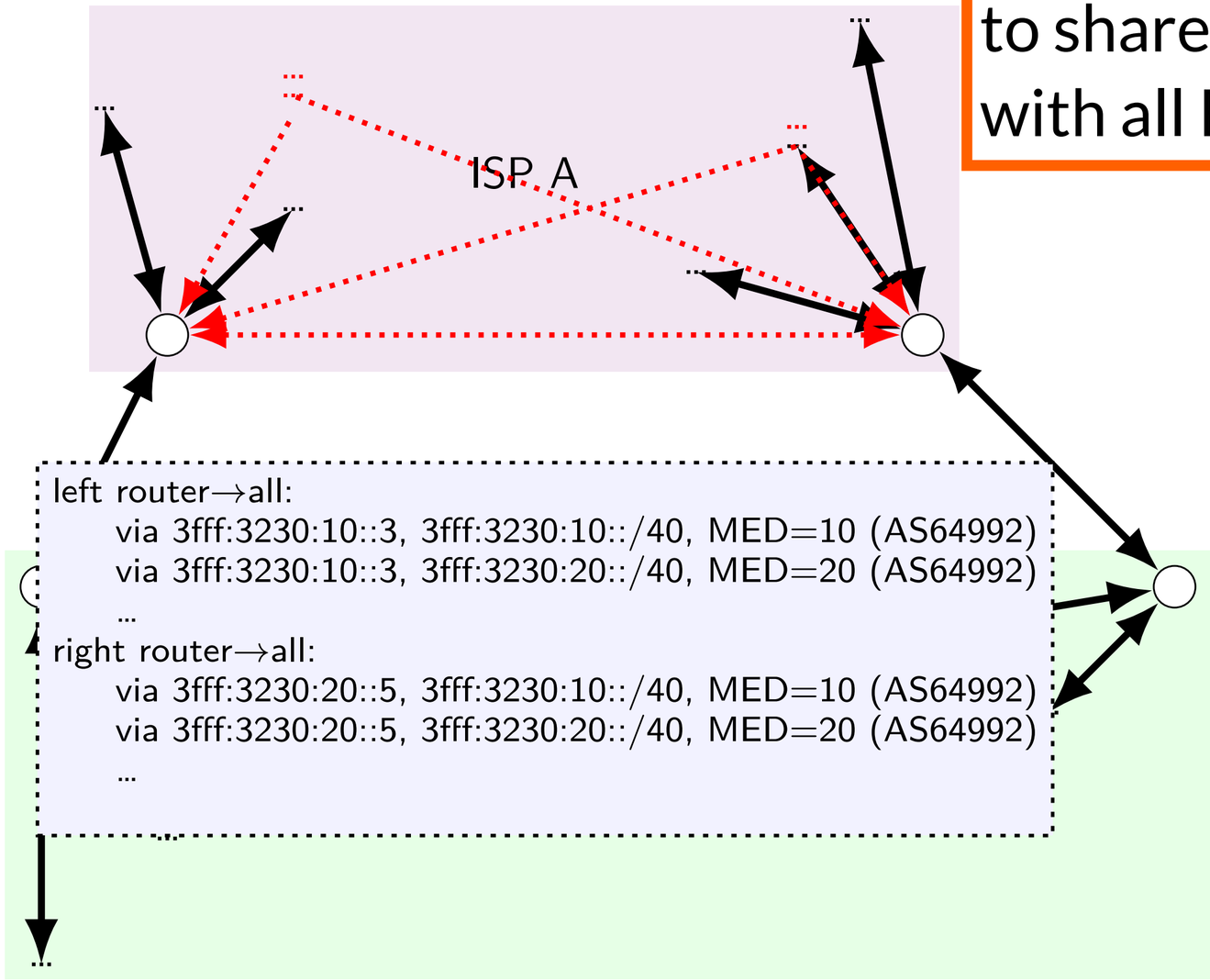
exchange possible routes on each connection

typically same routes for each connection to AS

but often different attributes

multiple BGP sessions

within ISP, use internal BGP (IBGP) to share everytig learned via BGP with all BGP routers



preference between routes

if multiple choices, most common strategy:…

should use most specific route

use 2001:db8:1234::/40 over 2001:db8:1234::/39 if both apply

(but usually reject very small address ranges (e.g. /31 for IPv4, /60 for IPv6))

then (if tie) local policy applies

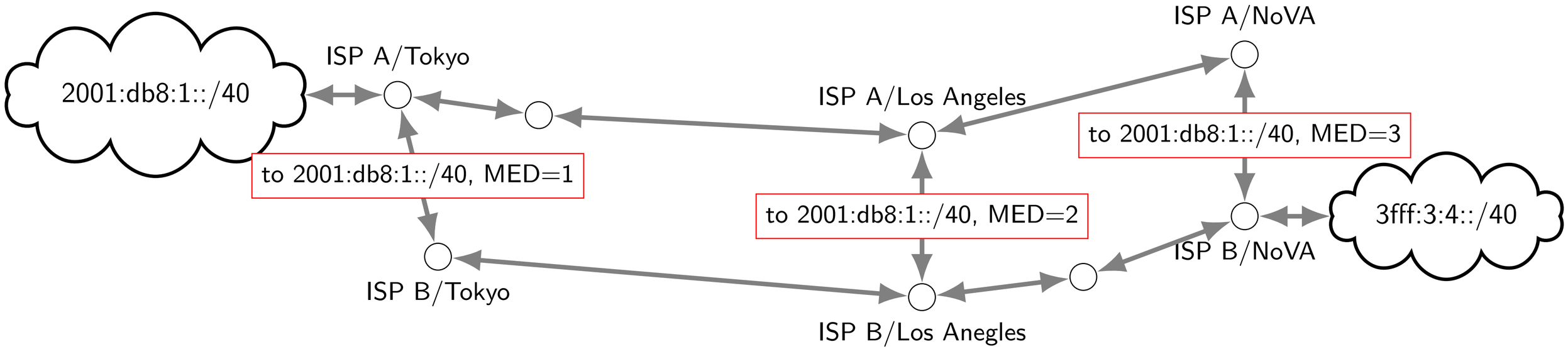
then shortest AS path

then lower AS number

then (sometimes) lower MED (multiple exit discriminator)

then best route within current AS

multi-exit discriminator



getting your preference

to affect how people route you, can…

prepend to AS path sent to make it longer

- typically add serveral copies of your AS number

only announce network from certain of your routers

- problem: won't have all 'backup' paths available

announce a large network in more specific pieces

- 3fff:1234::/32 as 3fff:1234::/33 and 3fff:1234:8000::/33

get other networks to change how they forward your routes

- often enabled through 'BGP communities'

https://bgp.he.net/super-lg/

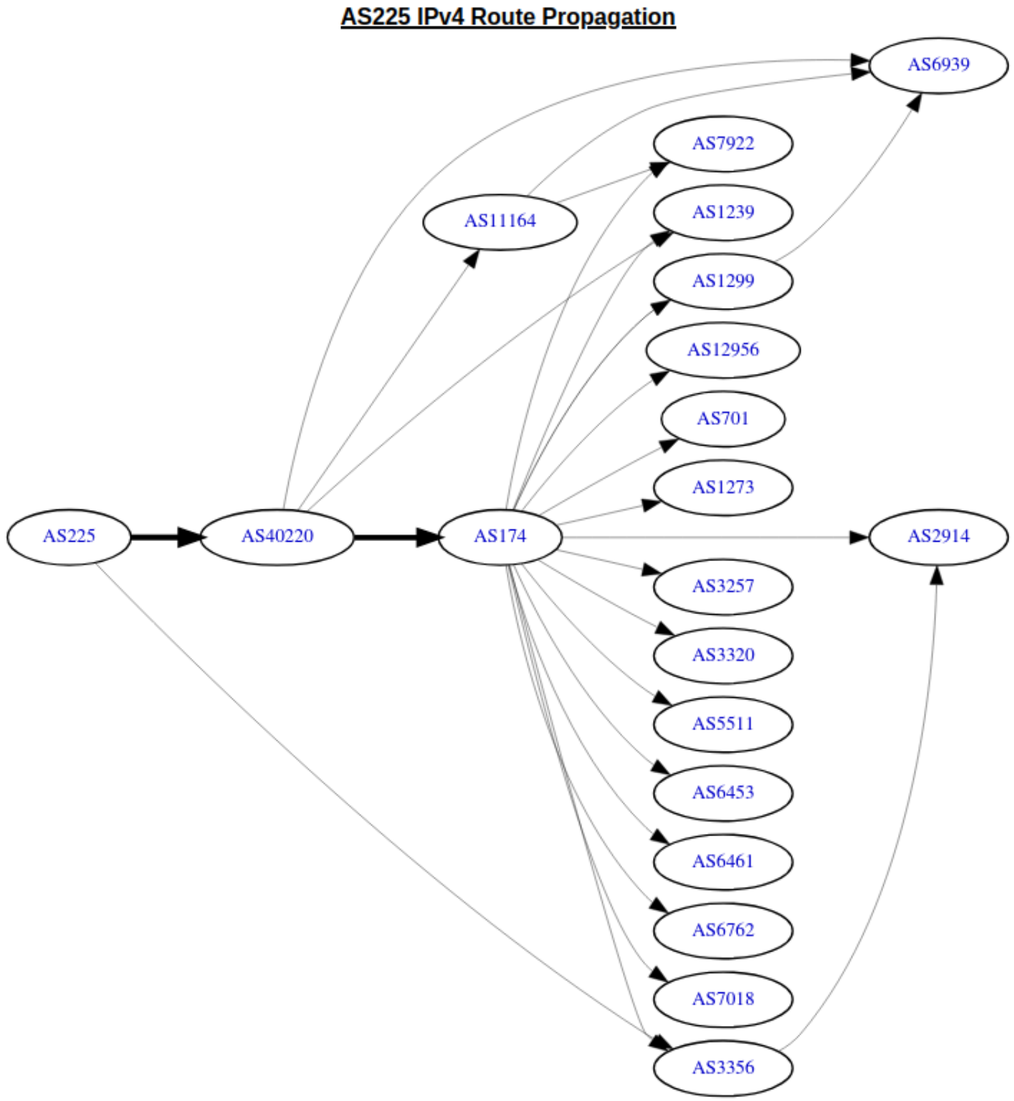
https://bgp.he.net/super-lg/#128.143.0.0/16?tob=none&mt=include&ma=6939&els=exact

https://lg.ring.nlnog.net/prefix?q=128.143.0.0/16&match=exact&peer=all

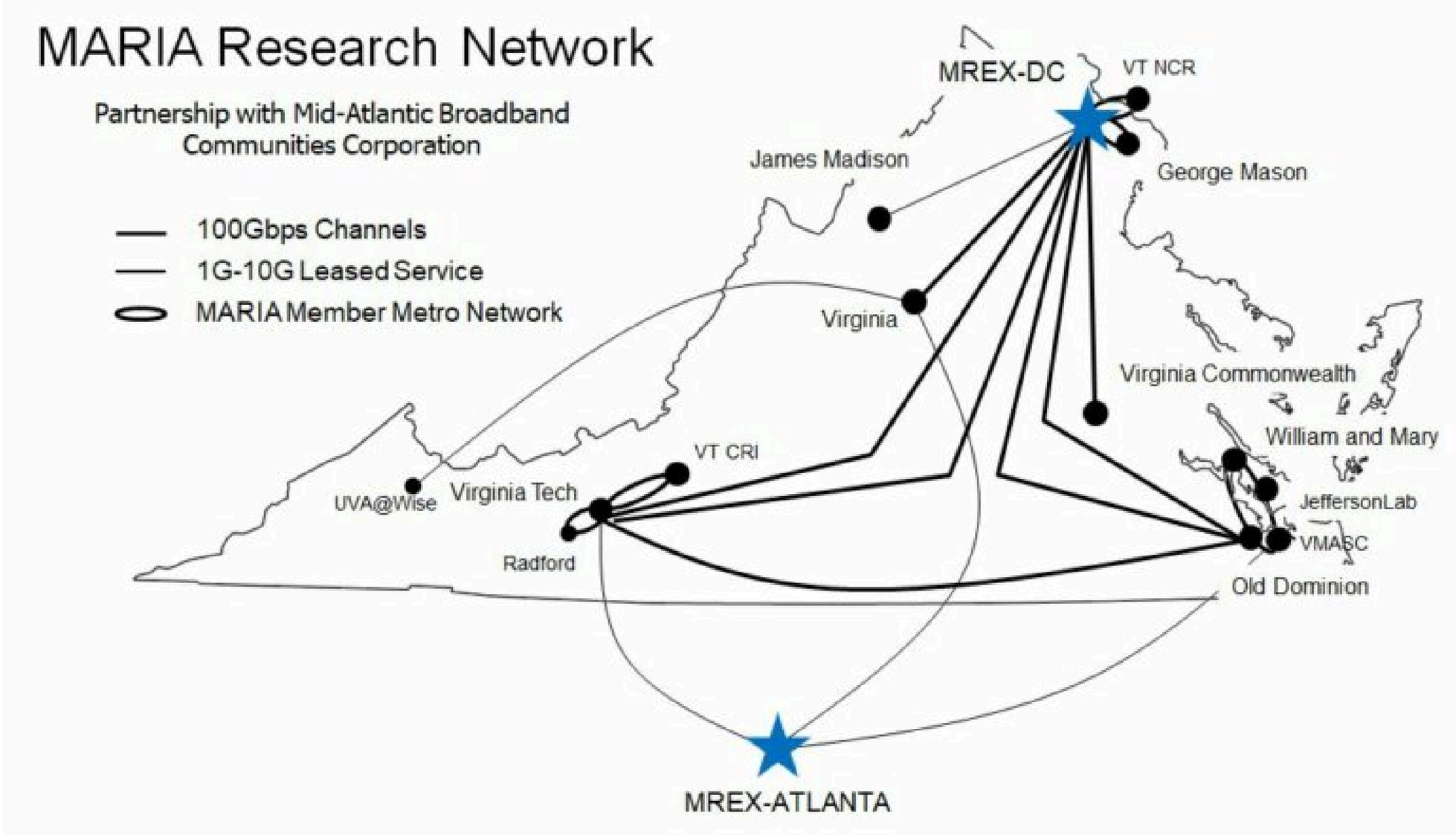
future slides mostly from 2024

UVa has changed their network configuration since then

https://bgp.he.net/AS225 (University of Virginia)



AS40220


MARIA Research Network
Partnership with Mid-Atlantic Broadband Communities Corporation
100Gbps Channels
1G-10G Leased Service
MARIA Member Metro Network
MREX-DC
VT NCR
James Madison
George Mason
Virginia
Virginia Commonwealth
William and Mary
VT CRI
UVA@Wise
Virginia Tech
JeffersonLab
VMASC
Radford
Old Dominion
MREX-ATLANTA

PeeringDB

Search here for a network, IX, or facility.

Advanced Search Legacy Search

Register or Login

English (English)

Mid-Atlantic Terascale Partnership - MATP

EXPORT

Organization	Mid-Atlantic Terascale Partnership - MATP
Also Known As	MARIA / Virginia Tech (Virginia Polytechnic Institute and State University)
Long Name	
Company Website	
ASN	40220
IRR as-set/route-set	AS40220:AS-ALL
Route Server URL	
Looking Glass URL	
Network Types	Educational/Research
IPv4 Prefixes	90
IPv6 Prefixes	25
Traffic Levels	Not Disclosed
Traffic Ratios	Mostly Inbound
Geographic Scope	Regional
Protocols Supported	✓ Unicast IPv4 ○ Multicast ✓ IPv6 ○ Never via route servers
Last Updated	2022-07-27T05:34:22Z
Public Peering Info Updated	2024-07-24T16:31:34Z
Peering Facility Info Updated	2016-03-14T20:59:43Z
Contact Info Updated	2020-01-22T04:24:10Z
Notes	MATP (AS40220) is a consortium of research institutions in Virginia, Maryland, and Washington formed to to support research activities that require next-generation high-performance network connectivity. We have routers in Equinix Ashburn and TelX Atlanta and provide commodity and R&E network services to most research universities in Virginia. MATP is managed by Virginia Tech (AS1312).

Public Peering Exchange Points

Filter

Exchange AŻ IPv4	ASN IPv6	Speed Port Location	RS Peer	BFD Support
Digital Realty Atlanta 206.126.110.182	40220 2001:504:17:110::182	10G	○	○
Equinix Ashburn 206.126.236.139	40220 2001:504:0:2:0:4:220:1	20G	○	○

Interconnection Facilities

Filter

Facility AŻ ASN	Country City
Digital Realty ATL (56 Marietta) 40220	United States of America Atlanta
Equinix DC1-DC15, DC21 - Ashburn 40220	United States of America Ashburn

AS3356

LUMEN®

SOLUTIONS PRODUCTS SHOP RESOURCES

DEDICATED INTERNET ACCESS

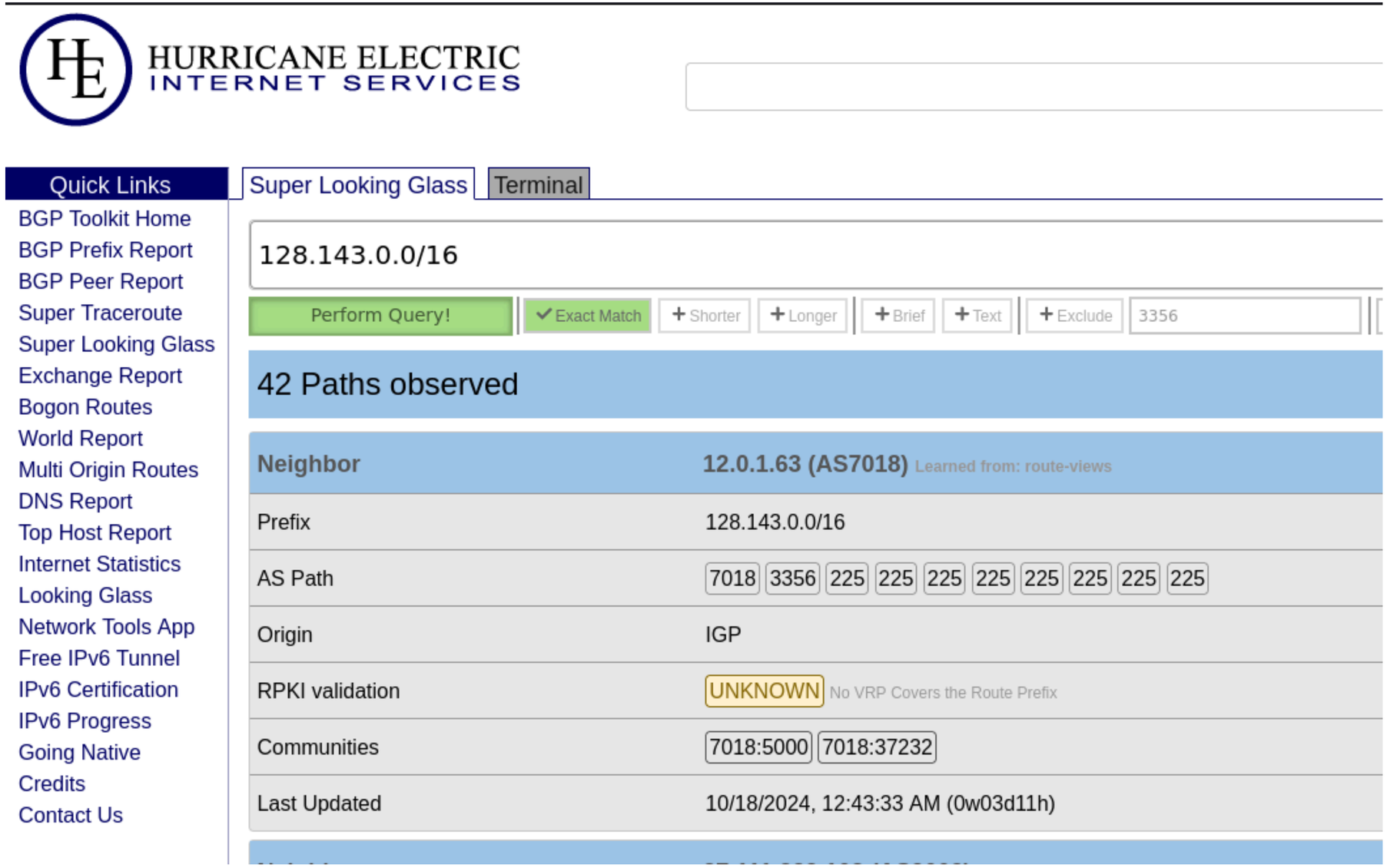
Go full speed

Give your enterprise a direct fast lane to 6,300+ interconnects[1] with speeds up to 100 Gbps.

GET OFFER VIEW SPECS

Overview Benefits Pricing Customer Stories Resources

2024: AS3356 is a backup (8x prepend)



2025: AS3356 and AS 40220 equal?

Neighbor	172.18.14.1 (AS6939) Learned from: collector14.bgp.he.net
Prefix	128.143.0.0/16
AS Path	6939 3356 225
Origin	IGP
RPKI validation	UNKNOWN No VRP Covers the Route Prefix
Last Updated	10/21/2025, 5:26:12 AM (0w00d08h)

Neighbor	172.18.14.2 (AS6939) Learned from: collector14.bgp.he.net
Prefix	128.143.0.0/16
AS Path	6939 40220 225
Origin	IGP
RPKI validation	UNKNOWN No VRP Covers the Route Prefix
Last Updated	9/30/2025, 10:19:43 AM (3w00d03h)

peeringdb

https://peeringdb.com — commonly used database of ASes and how to peer with them

there is also – “whois” records (from RIRs) for ASes, IP blocks with contact info

internet exchanges and route servers

internet exchange

- local network (typically within metro area) for connecting networks
- often run at and/or by ‘carrier-neutral’ datacenter
- typically high bandwidth (10-100Gbps ports to network)

route servers

- BGP servers run by internet exchange
- consolidates routes from participants
- goal: only need O(n) BGP connections, not O(n^2)

BGP communities

routes sent via BGP can have ‘communities’

extra information tagged on routes sent via BGP

large ISPs have lists of communities their customers/peers can use

… and these affect how those routes are used

aside: Internet2

non-profit networking consortium

operations major US University-focused network

also makes eduroam work across different Universities

one of MARIA's major sources of connectivity (and indirectly one of UVA's major sources of connectivity)

has lots of peering relationships (incl. with big Internet companies)

(but not general Internet provider)

selected Internet2 BGP communities

From Internet2 Documentation:

Internet2 External Traffic Influencing Communities

International, Non-International, or FEDNET peers may send the below community and we will set their localpref to 460 or 560 respectively:

Default - local-pref 500

11537:40 - Low (local-pref 460)

11537:160 - High (local-pref 560)

Connectors may send the below community and we will set their localpref to 540 or 620 respectively:

Default - local-pref 600

11537:140 - Low (local-pref 540)

11537:260 - High (local-pref 620)

Internet2 Peers may send the following communities:

11537:2002 - Block prefix to commercial R&E peers.

Internet2 International (ITN) peers may send the below communities for path prepending:

65001:65000 - prepend x1

65002:65000 - prepend x2

65003:65000 - prepend x3

Internet2 Internal Traffic Influencing Communities

Internet2-connected Gigapops and connectors with heterogeneous participants (i.e. not all are universities) may need to use these communities to help them pass prefixes appropriately to their participants. For example, connectors can pass all prefixes on to their university participants, but should not pass commercial or US Fednet prefixes to their commercial or Fednet participants.

11537:888 - Block to external (Internet2 uses this community to mark prefixes that will not be reannounced to external peers).

11537:902 - non-I2 R&E sites sponsored for connection by members.

11537:910 - sponsored educational groups (Primary state networks).

11537:950 - Internet2 Participants.

11537:2000 - Commercial research - Lab participant.

11537:2001 - Special Commercial prefix used for V6 or multicast.

11537:2500 - Block advertisement to international.

11537:2501 - International.

....

NET+ BGP Communities

Internet2 has implemented a set of BGP communities that, along with your own local BGP import policy, will allow you to control your traffic path to/from the Internet2 NET+ peer networks. All of these communities are specific to individual peer networks and do not affect routing to/from the other peer networks.

11537:5000 - Internet2 NET+ Services.

11537:5002 - Cloud.

....

The following community combination of <CODE>:<ASN> allows you to block or prepend prefixes sent to individual NET+ peers.

Codes:

- 65000 - prefixes will not be sent to NET+ peer's ASN
- 65001 - prefixes will be prepended 1 time to NET+ peer's ASN
- 65002 - prefixes will be prepended 2 times to NET+ peer's ASN
- 65003 - prefixes will be prepended 3 time2 to NET+ peer's ASN

NET+ peer ASN's

- 16509 - Amazon
- 62715 - Code42
- ...

community options from prev slide

setting local-pref:

- you can decide how preferred your route is by Internet2
- maybe to make one primary, another secondary?

blocking route from being sent to specific place

prepending Internet2's AS before forwarding prefix

- hopefully make that route less preferred by others

prepending Internet2's AS before forwarding prefix to specific place

- hopefully make that route less preferred by that place

other things with communities

Internet2 also uses communities to mark…

what location routes were learned from

what type of organization routes were learned from

whether Internet2 is only allowed to use the route non-commerically or not

…

AS7007

https://seclists.org/nanog/1997/Apr/444

7007 Explanation and Apology

From: “Vincent J. Bono” <vbono () MAI NET>
Date: Sat, 26 Apr 1997 19:41:35 EST

```
Dear All,

    I would like to sincerely apologize to everyone everwhere who
experienced problems yesterday due to the 7007 AS announcements.

    If anyone cares to know, here is what happened:

    At 11:30AM, EST, on 25 Apr 1997, our border router, stamped with
AS 7007, recieved a full routing view from a downstream ISP (well, a
view contacing 23,000 routes anyway).

    There was no distibute list imposed on the downstream since they
also advertise their customer AS's to us (they were also
experimenting with sending some routes out through us and some out
through the MAE).  We did filter out routes from them containing any
of our AS numbers but since they got the view from someone at
MAE-East none of our internal AS numbers showed up at all.  Not
having a filter imposed on the inbound side was our error.

    In an as yet unexplained twist of bits, the 7007 router then
began to de-aggregate the 23K route view *and* strip the AS path out
of it.  I will emphasize that we were running no IGP at the time.
Not one.  Not OSPF, not RIP, nothing.
```

...

2008 Pakistan Youtube

Pakistan Telecom recieved gov’t order to block youtube

implemented by inserting route for YouTube’s IP in internal network

misconfiguration meant route was advertised on BGP

was more specific than YouTube’s route, so made YouTube unreachablej

timeline from RIPE NCC

https://www.ripe.net/about-us/news/youtube-hijacking-a-ripe-ncc-ris-case-study/

Youtube is announcing 208.65.152.0/22

18:47Z: Pakistan Telecom starts announcing 208.65.153.0/24

20:07Z: Youtube starts announcing 208.65.153.0/24

20:18Z: Youtube starts announcing 208.65.153.0/25 and 208.65.153.128/25

20:51Z: Pakistan Telecom's ISP forwards their announcements with additional copy of Pakistan Telecom's AS number

21:01Z: Pakistan Telecom's ISP withdraws routes initiated by Pakistan Telecom (but not Pakistan Telecom's customers)

BGP Hijacking targeted cryptocurrency stuff

KLAYswap (Feb 2022), Celer Bridge (Sep 2022)

attackers intentionally redirected traffic to malicious version of services

… and stole money

both probably spoofed the final AS number in AS path

sometimes involved adding attacked IP range to routing registry

nation-states?



a more recent example

ars TECHNICA AI BIZ & IT CARS CULTURE GAMING HEALTH POLICY SCIENCE SECURITY SPACE TECH FORUM | SUBSCRIBE

HACKING INTERNET ROUTING

BGP hijack infecting networks caused by a comedy of errors that's not funny at all

What can we learn from a BGP hijacking that poisoned production software? Plenty.

DAN GOODIN – SEP 2, 2026 7:00 AM | 68



Credit: Getty Images

route security

historically, no verification routes announced by "owner" of IP addresses

Internet Routing Registries — database of AS to IP address

- used automatically to filter out mistakes
- (not really designed to resist malicious attacks)

some verify which IPs they should have with RIRs

- "letter of agency/authority" to delegate

effort to deploy RPKI — public-key based scheme to verify routes

- checks that routes originated at correct AS
- doesn't verify intermediate ASes will forward correctly

Internet Engineering Task Force (IETF)
Request for Comments: 6480
Category: Informational
ISSN: 2070-1721

M. Lepinski
S. Kent
BBN Technologies

February 2012

An Infrastructure to Support Secure Internet Routing

Abstract

This document describes an architecture for an infrastructure to support improved security of Internet routing. The foundation of this architecture is a Resource Public Key Infrastructure (RPKI) that represents the allocation hierarchy of IP address space and Autonomous System (AS) numbers; and a distributed repository system for storing and disseminating the data objects that comprise the RPKI, as well as other signed objects necessary for improved routing security. As an initial application of this architecture, the document describes how a legitimate holder of IP address space can explicitly and verifiably authorize one or more ASes to originate routes to that address space. Such verifiable authorizations could be used, for example, to more securely construct BGP route filters.

partial tables

dealing with full Internet routing table is expensive

common shortcut if you have a couple ISPs:

keep ‘short’ routes (example: short AS path)

- to one of your “primary” ISPs
- maybe using ECMP

have default route for other cases

- special route for 0.0.0.0/0, ::/0
- to one of your ISPs

take advantage of more specific routes beating less specific

- typically also true in OSPF, RIP, etc.

backup slides